Advanced Strategy Analysis: Funding Rate Arbitrage

Strategy: Perpetual DEX funding rate arbitrage

Platforms: dYdX / Hyperliquid × Binance spot

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1. How It Works

Perpetual futures contracts have no expiry date. To keep the perp price anchored to spot, exchanges pay a **funding rate** every 8 hours: if the perp trades above spot, longs pay shorts; if below spot, shorts pay longs.

The arb: When funding rate is strongly positive, you can:

1. **Short** the perp on Hyperliquid (collect funding every 8h)
2. **Buy** the same asset on Binance spot (hedge the directional exposure)

You are market-neutral (delta = 0) but collect funding continuously.

Example with ETH at $2,000, funding rate = +0.05%/8h (annualised ~54%):

Position:

  Long $1,000 ETH on Binance spot

  Short $1,000 ETH perp on Hyperliquid

Every 8 hours:

  Funding received = $1,000 × 0.05% = $0.50

  Per day (3 payments): $1.50

  Per year: ~$547 on $2,000 capital = 27.35% APY

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2. Capital Requirements

- **Minimum viable:** ~$500 per side ($500 spot + $500 perp margin)
- **Margin requirement:** Hyperliquid requires ~10% initial margin → 10× leverage cap

- **Practical:** Use 1–2× leverage to avoid liquidation risk

Capital	Position size	Daily income (0.05% rate)
$500/side	$500	~$1.50
$5,000/side	$5,000	~$15
$50,000/side	$50,000	~$150

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3. Risks

Risk	Description	Mitigation
Funding rate flip	Rate turns negative → you pay instead of receive	Monitor rate, close position when < 0.01%
Liquidation	Perp moves far against you before spot hedge rebalances	Use low leverage (1–2×), set stop-loss
Exchange risk	Hyperliquid smart contract exploit or dYdX insolvency	Cap exposure per venue, use established platforms
Basis risk	Spot and perp prices diverge temporarily	Harmless if held to funding payment, matters if forced to close early
Gas costs	Frequent rebalancing on-chain	Batch rebalances, use Arbitrum for low gas

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4. Implementation Connection

Modules reusable from this project:

Existing module	Reuse
executor/engine.py	State machine for two-leg execution (long spot + short perp)

strategy/fees.py	Extend FeeStructure to include funding rate income
safety/risk.py	RiskManager, RiskLimits unchanged
monitoring/telegram.py	Alerts for funding rate changes
monitoring/metrics.py	Add FUNDING_RATE gauge, FUNDING_PNL counter

New code needed:

- exchange/hyperliquid_client.py — REST/WS client for Hyperliquid API
- strategy/funding_rate.py — FundingRateSignal generator (checks rate every 8h)
- executor/perp_executor.py — opens/closes perp position
- Background loop: monitor rate, auto-close when rate < threshold

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5. Expected Returns

Scenario	Funding rate	APY on $1,000/side
Conservative	0.01%/8h = 10.9% APR	~11%
Normal	0.03%/8h = 32.8% APR	~33%
Good	0.05%/8h = 54.7% APR	~55%
Bull market peak	0.1%/8h = 109% APR	~109%

Realistic expectation: Rates average 0.02–0.04%/8h over a year, giving **~22–44% APY** on deployed capital after gas costs. Significantly more predictable than CEX/DEX spot arb since you capture income rather than hunting spread windows.

Why I would pursue it after the internship:

Well, first of all it is more predictable and has a stable growth, unlike arbitrage on tokens - where you need a lot of setup to beat other bots and the token can grow or decline, which makes it profitable on larger amounts with good arbitrage payouts, but still first - you need to get those money and the inventory to do it, or a setup so that you can catch good opportunities.

With this approach tho, you can have 500/500 on short and long positions that cancel each other out, if it falls I win short if it grows - I win long, and you also get a funding income - that is avg 0.03% over the year, so it comes to 16% increase of inventory per year which is nice. For sure it has risks, and you need to manage that really carefully, same as with arbitrage, but there are just less points of failure and less gambling on will the coin grow or fall - losing us inventory.